

Computer Networks 1

Wireshark Lab: Getting Started

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Câu 1. List 3 different protocols that appear in the protocol column in the unfiltered packet-listing window: TCP, DNS, HTTP

Câu 2. How long did it take from when the HTTP GET message was sent until the HTTP OK reply was received?

Message	Package No.	Time (s)
HTTP GET	277	2026-10-02 09:07:55.285997600
HTTP/1.1 200 OK	281	2026-10-02 09:07:55.620423900

$\Delta t = 2026-10-02\ 09:07:55.620423900 - 2026-10-02\ 09:07:55.285997600 = 0.334426300\text{ s} = 334.426300\text{ ms}$

Ở gói OK, Time since request: 334.426300 milliseconds -> khớp kết quả tính.

Câu 3. Internet Address (IP Address)

	IP Address
gaia.cs.umass.edu	128.119.245.12
My computer	172.20.10.3

Package No. 277: Source = 172.20.10.3 (My computer)

Destination = 128.119.245.12 (gaia.cs.umass.edu)

Câu 4. Message (GET & OK)

a) HTTP GET message:

No.	Time	Source	Destination	Protocol	Length	Info
277	2026-10-02 09:07:55.285997600	172.20.10.3	128.119.245.12	HTTP	562	GET /wireshark-labs/INTRO-wireshark-file1.html HTTP/1.1

Frame 277: Packet, 562 bytes on wire (4496 bits), 562 bytes captured (4496 bits) on interface \Device\NPF_{27A288C2-5E61-46C7-88FD-AE48BEC7B0FA}, id 0

Ethernet II, Src: ChongqingFug_62:65:25 (8c:c8:4b:62:65:25), Dst: fa:4e:73:4d:d6:64 (fa:4e:73:4d:d6:64)

Internet Protocol Version 4, Src: 172.20.10.3, Dst: 128.119.245.12

Transmission Control Protocol, Src Port: 50312, Dst Port: 80, Seq: 1, Ack: 1, Len: 508

Hypertext Transfer Protocol

GET /wireshark-labs/INTRO-wireshark-file1.html HTTP/1.1\r\n

Request Method: GET

Request URI: /wireshark-labs/INTRO-wireshark-file1.html

Request Version: HTTP/1.1

Host: gaia.cs.umass.edu\r\n

Connection: keep-alive\r\n

Upgrade-Insecure-Requests: 1\r\n

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/154.0.0.0 Safari/537.36 Edg/154.0.0.0\r\n

Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7\r\n

Accept-Encoding: gzip, deflate\r\n

Accept-Language: vi,en;q=0.9,en-GB;q=0.8,en-US;q=0.7\r\n

\r\n

[Response in frame: 281]

[Full request URI: http://gaia.cs.umass.edu/wireshark-labs/INTRO-wireshark-file1.html]

b) HTTP OK message:

No.	Time	Source	Destination	Protocol	Length	Info
281	2026-10-02 09:07:55.620423900	128.119.245.12	172.20.10.3	HTTP	495	HTTP/1.1 200 OK (text/html)

Frame 281: Packet, 495 bytes on wire (3960 bits), 495 bytes captured (3960 bits) on interface \Device\NPF_{27A288C2-5E61-46C7-88FD-AE48BEC7B0FA}, id 0

Ethernet II, Src: fa:4e:73:4d:d6:64 (fa:4e:73:4d:d6:64), Dst: ChongqingFug_62:65:25 (8c:c8:4b:62:65:25)

Internet Protocol Version 4, Src: 128.119.245.12, Dst: 172.20.10.3

Transmission Control Protocol, Src Port: 80, Dst Port: 50312, Seq: 1, Ack: 509, Len: 441

Hypertext Transfer Protocol

HTTP/1.1 200 OK\r\n

Response Version: HTTP/1.1

Status Code: 200

[Status Code Description: OK]

Response Phrase: OK

Date: Fri, 02 Oct 2026 02:07:55 GMT\r\n

Server: Apache/2.4.62 (AlmaLinux) OpenSSL/3.5.5 mod_fcgid/2.3.9 mod_perl/2.0.12 Perl/v5.32.1\r\n

Last-Modified: Tue, 28 Oct 2025 05:59:01 GMT\r\n

ETag: "51-64231b6715777"\r\n

Accept-Ranges: bytes\r\n

Content-Length: 81\r\n

Keep-Alive: timeout=5, max=100\r\n

Connection: Keep-Alive\r\n

Content-Type: text/html; charset=UTF-8\r\n

\r\n

[Request in frame: 277]

[Time since request: 334.426300 milliseconds]

[Request URI: /wireshark-labs/INTRO-wireshark-file1.html]

[Full request URI: http://gaia.cs.umass.edu/wireshark-labs/INTRO-wireshark-file1.html]

File Data: 81 bytes

Line-based text data: text/html (3 lines)